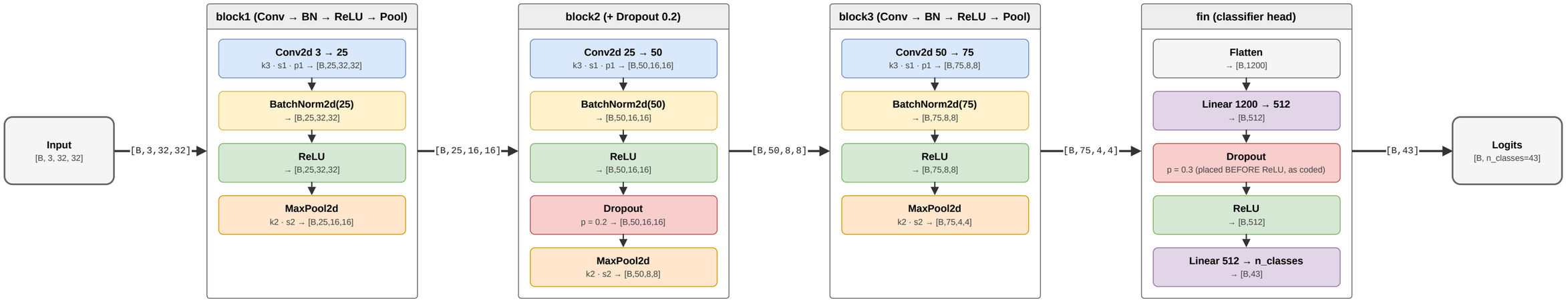
TrialCNN — custom 3-block CNN · drawn for IMG_SIZE=32, n_classes=43 (flat_sz = 75 × (IMG_SIZE//8)²)

Input / Output / Reshape | Conv | Depthwise Conv | BatchNorm | Activation | Pooling | Dropout / StochDepth | Linear | SE branch | Composite block

red dashed line = skip / residual path



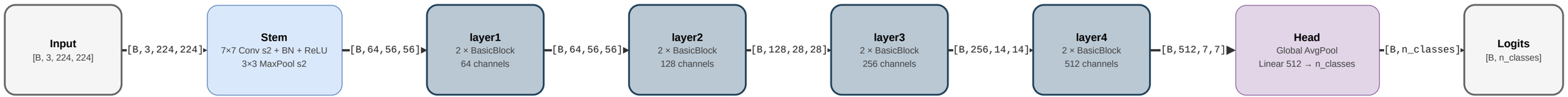
Notes

- flat_sz = 75 × (IMG_SIZE//8)² = 1200 for IMG_SIZE=32 — update the Flatten / Linear labels if your config differs.
- Three 2×2 max-pools → spatial size /8. Every conv is 3×3, stride 1, padding 1 (size-preserving).
- Trainable parameters ≈ 683,096 (IMG_SIZE=32, n_classes=43).

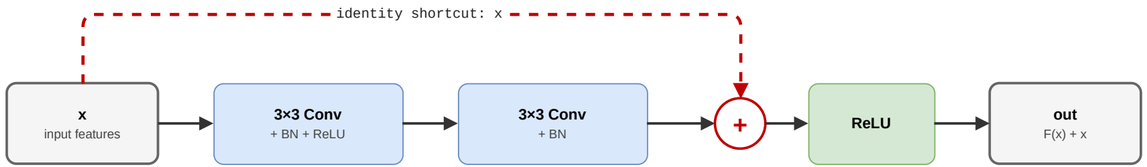
ResNet-18 — residual learning · input 224×224, output n_classes logits

Input / Output | Conv (+ BN, activation) | Group of blocks | Pool + classifier | **red dashed line = skip connection**

1. Overall architecture



2. Inside one BasicBlock (the repeating unit)



Key points for viva
- The block learns a residual F(x) and outputs F(x) + x, so gradients flow through the skip path and deep nets train easily.
- 4 stages (64 → 128 → 256 → 512 channels); each stage after the first halves the resolution (stride 2) and uses a 1×1 conv on the skip path to match shapes.
- 8 BasicBlocks × 2 convs + stem conv + FC = 18 weight layers · ≈ 11.2 M parameters.

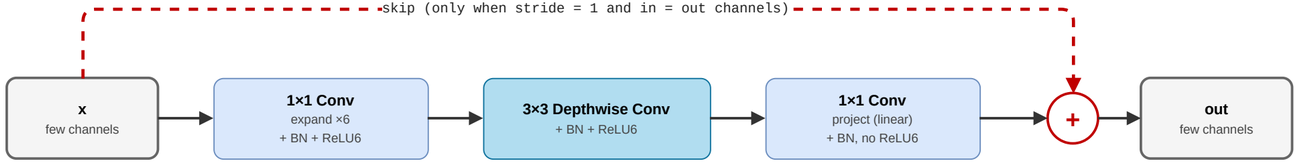
MobileNetV2 — lightweight · input 224×224, output n_classes logits

Input / Output | Conv (+ BN, activation) | Group of blocks | Depthwise Conv | Pool + classifier | **red dashed line = skip connection**

1. Overall architecture



2. Inside one Inverted Residual block (narrow → wide → narrow)



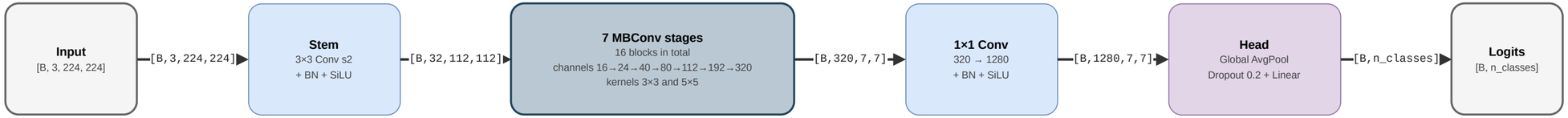
Key points for viva

- Depthwise separable convolution (3×3 depthwise + 1×1 pointwise) gives far fewer parameters and FLOPs than a normal conv.
- Inverted residual: the skip connects the thin (bottleneck) ends, and the middle is expanded ×6; the last 1×1 is linear so information is not lost.
- ≈ 2.3 M parameters with n_classes = 43.

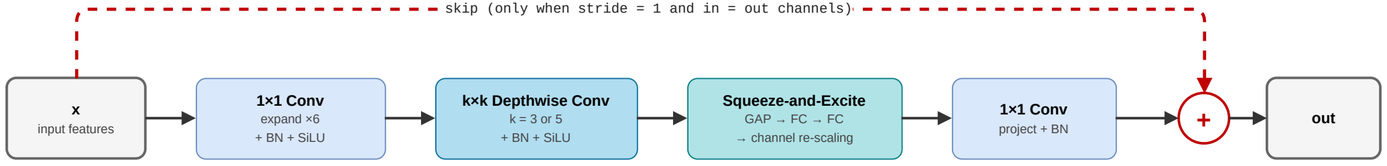
EfficientNet-B0 — compound-scaled baseline · input 224×224, output n_classes logits

Input / Output | Conv (+ BN, activation) | Group of blocks | Depthwise Conv | Squeeze-and-Excitation | Pool + classifier | red dashed line = skip connection

1. Overall architecture



2. Inside one MBConv block



Key points for viva
- Compound scaling: depth, width and resolution are scaled together with one coefficient; B0 is the baseline network found by NAS.
- MBConv = MobileNetV2 inverted residual + Squeeze-and-Excitation (learns which channels matter) + SiLU activation.
- ≈ 4.1 M parameters with n_classes = 43.